Separation of Variables for Three Model PDEs

Trig vs. Complex Exponential Approaches

Fourier Coefficients and Smoothness

Let $f(x)$ be π-periodic with Fourier series

$$f(x) = \sum_{n=1}^{\infty} \left(a_n \cos(nx) + b_n \sin(nx)\right).$$

- If $f \in C^k$, then $|a_n|, |b_n| = O(n^{-k-1})$
- If f is C^∞, coefficients decay faster than any power
- If f is analytic, coefficients decay exponentially

Time evolution in PDEs acts directly on Fourier modes.

Example: Twice Differentiable Function

Consider the 2π-periodic function on $0 < x < \pi$:

$$f(x) = x^2(\pi - x), \quad 0 < x < \pi$$

- Smooth: $f \in C^2$, but $f'''(x)$ is discontinuous at $x = 0, \pi$
- Odd extension (for sine series):

$$f(-x) = -f(x)$$

Sine series expansion:

$$f(x) = \sum_{n=1}^{\infty} b_n \sin(nx), \quad b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin(nx)\, dx$$

Decay rate: Since $f \in C^2$, $|b_n| = O(n^{-3})$

- Slower decay than infinitely smooth functions
- Smoother functions have faster Fourier coefficient decay

Problem Setup

Consider $0 < x < \pi$, $t > 0$, with periodic boundary conditions:

$$u(0,t) = u(\pi,t), \quad u_x(0,t) = u_x(\pi,t).$$

We compare:

- **Linear advection:** $u_t = u_x$
- **Heat equation:** $u_t = u_{xx}$
- **Wave equation:** $u_{tt} = u_{xx}$

Trig Separation Ansatz

Assume

$$u(x,t) = X(x)T(t),$$

with periodic spatial eigenfunctions:

$$X_n(x) = \cos(nx), \quad \sin(nx), \quad n \in \mathbb{N}.$$

Linear Advection: $u_t = u_x$

Substitute $u = X(x)T(t)$:

$$\frac{T'}{T} = \frac{X'}{X} = \lambda.$$

For $X_n(x) = \cos(nx), \sin(nx)$:

$$\lambda = \pm in.$$

$$T_n(t) = e^{\pm int}.$$

$$u(x,t) = \sum_n \left[a_n \cos(n(x+t)) + b_n \sin(n(x+t))\right].$$

Modes translate without decay.

Heat Equation: $u_t = u_{xx}$

Separation gives

$$\frac{T'}{T} = \frac{X''}{X} = -n^2.$$

$$T_n(t) = e^{-n^2 t}.$$

$$u(x,t) = \sum_n e^{-n^2 t} \left[a_n \cos(nx) + b_n \sin(nx) \right].$$

High frequencies decay rapidly.

Wave Equation: $u_{tt} = u_{xx}$

Separation yields

$$\frac{T''}{T} = \frac{X''}{X} = -n^2.$$

$$T_n(t) = A_n \cos(nt) + B_n \sin(nt).$$

$$u(x,t) = \sum_n \left[a_n \cos(nx) \cos(nt) + b_n \sin(nx) \sin(nt) \right].$$

Energy preserved; oscillatory behavior.

Complex Exponential Basis

Use

$$X_k(x) = e^{ikx}, \quad k \in \mathbb{Z}.$$

Periodic boundary conditions are automatic.

$$u(x,t) = \sum_{k \in \mathbb{Z}} \hat{u}_k(t) e^{ikx}.$$

Linear Advection: Complex Form

Substitute into $u_t = u_x$:

$$\hat{u}_k'(t) = ik\hat{u}_k(t).$$

$$\hat{u}_k(t) = \hat{u}_k(0)e^{ikt}.$$

$$u(x,t) = \sum_k \hat{u}_k(0)e^{ik(x+t)}.$$

Pure phase shift in time.

Heat Equation: Complex Form

From $u_t = u_{xx}$:

$$\hat{u}_k'(t) = -k^2 \hat{u}_k(t).$$

$$\hat{u}_k(t) = \hat{u}_k(0) e^{-k^2 t}.$$

Exponential damping of high frequencies.

Wave Equation: Complex Form

From $u_{tt} = u_{xx}$:

$$\hat{u}_k''(t) + k^2 \hat{u}_k(t) = 0.$$

$$\hat{u}_k(t) = A_k e^{ikt} + B_k e^{-ikt}.$$

Left- and right-traveling waves.

Characteristic Solution: Linear Advection

For

$$u_t = u_x,$$

characteristics satisfy

$$\frac{dx}{dt} = -1.$$

$$x + t = \text{constant}.$$

$$\boxed{u(x,t) = f(x+t)}$$

Initial profile transported without distortion.

Characteristic Solution: Wave Equation

For

$$u_{tt} = u_{xx},$$

define characteristic variables

$$\xi = x + t, \quad \eta = x - t.$$

General solution:

$$\boxed{u(x,t) = f(x+t) + g(x-t)}$$

Superposition of left- and right-moving waves.

Summary

Equation	Mode Behavior	Long-Time Effect
Advection	Phase shift	Translation
Heat	Exponential decay	Smoothing
Wave	Oscillation	Energy conservation